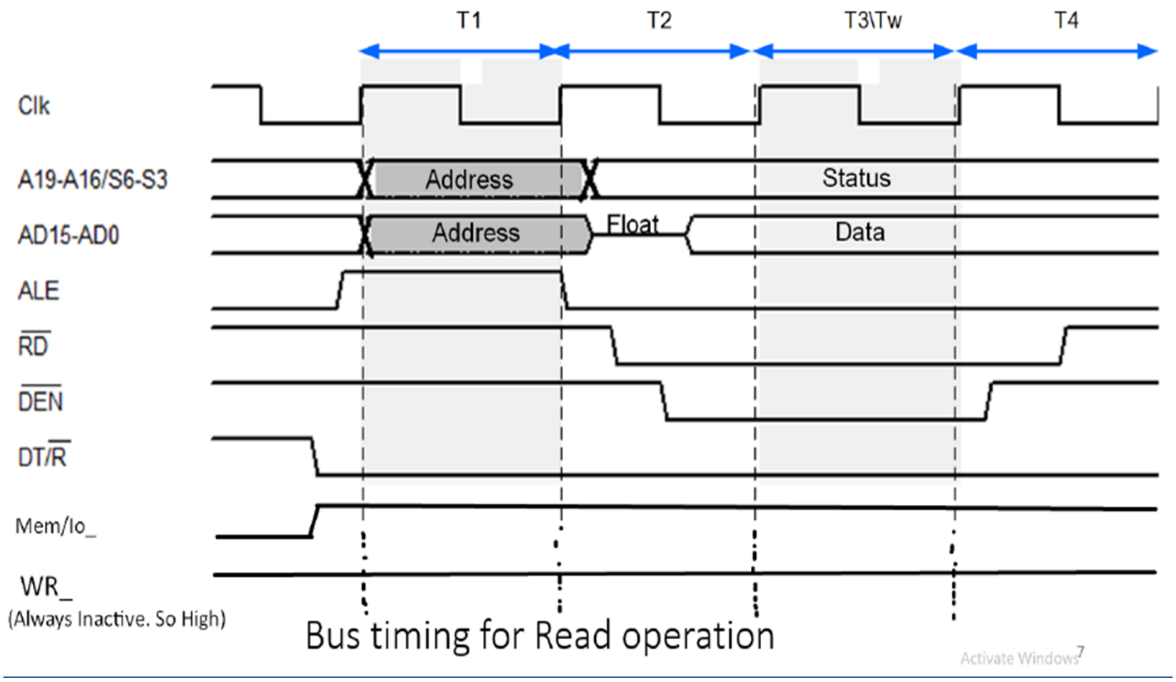
Solution of Assignment - 02

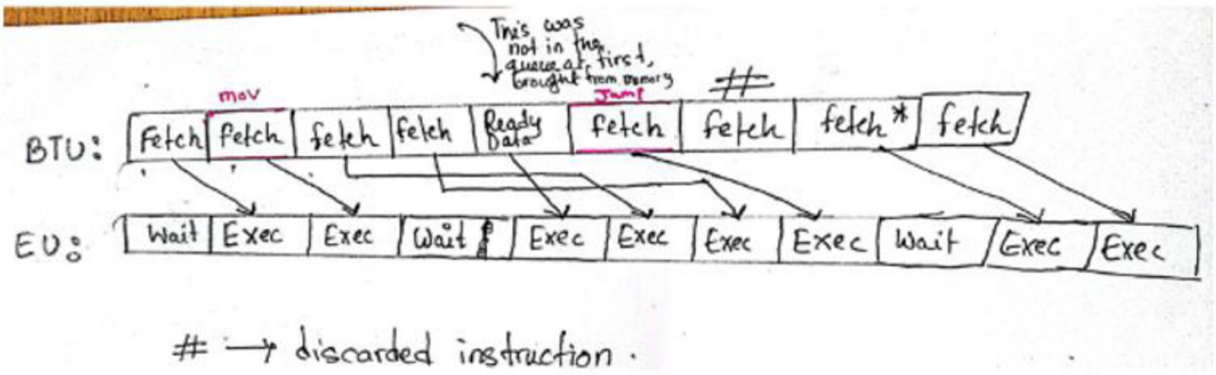
1.



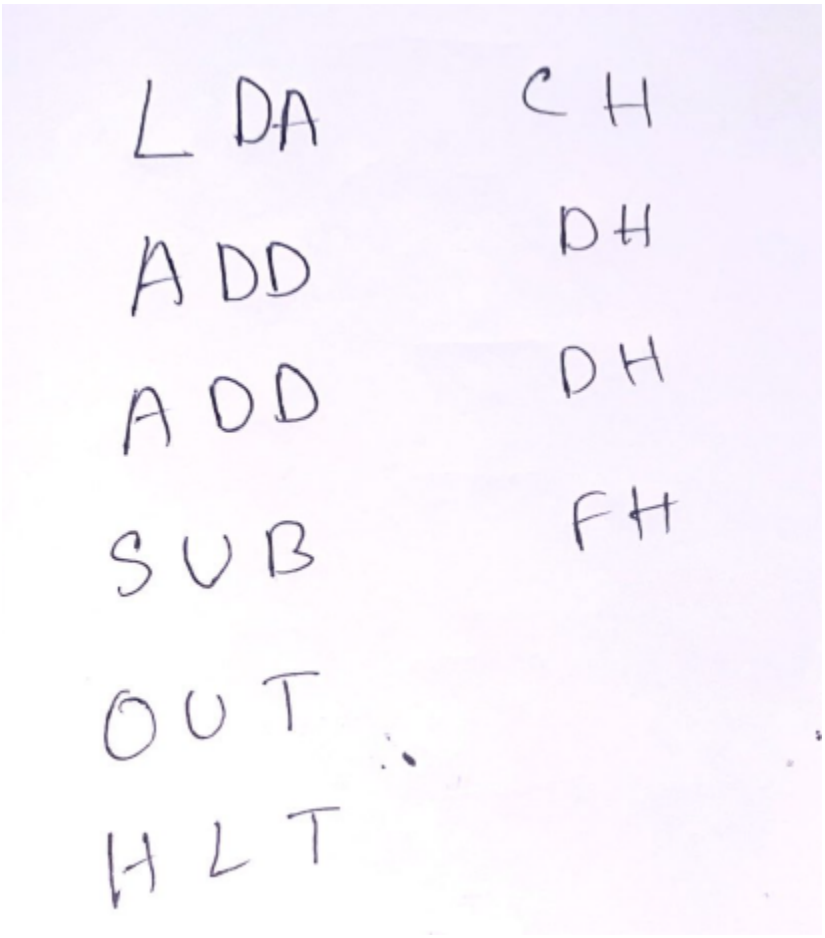
Bus timing for Read operation

Activate Windows

2.



3. a)

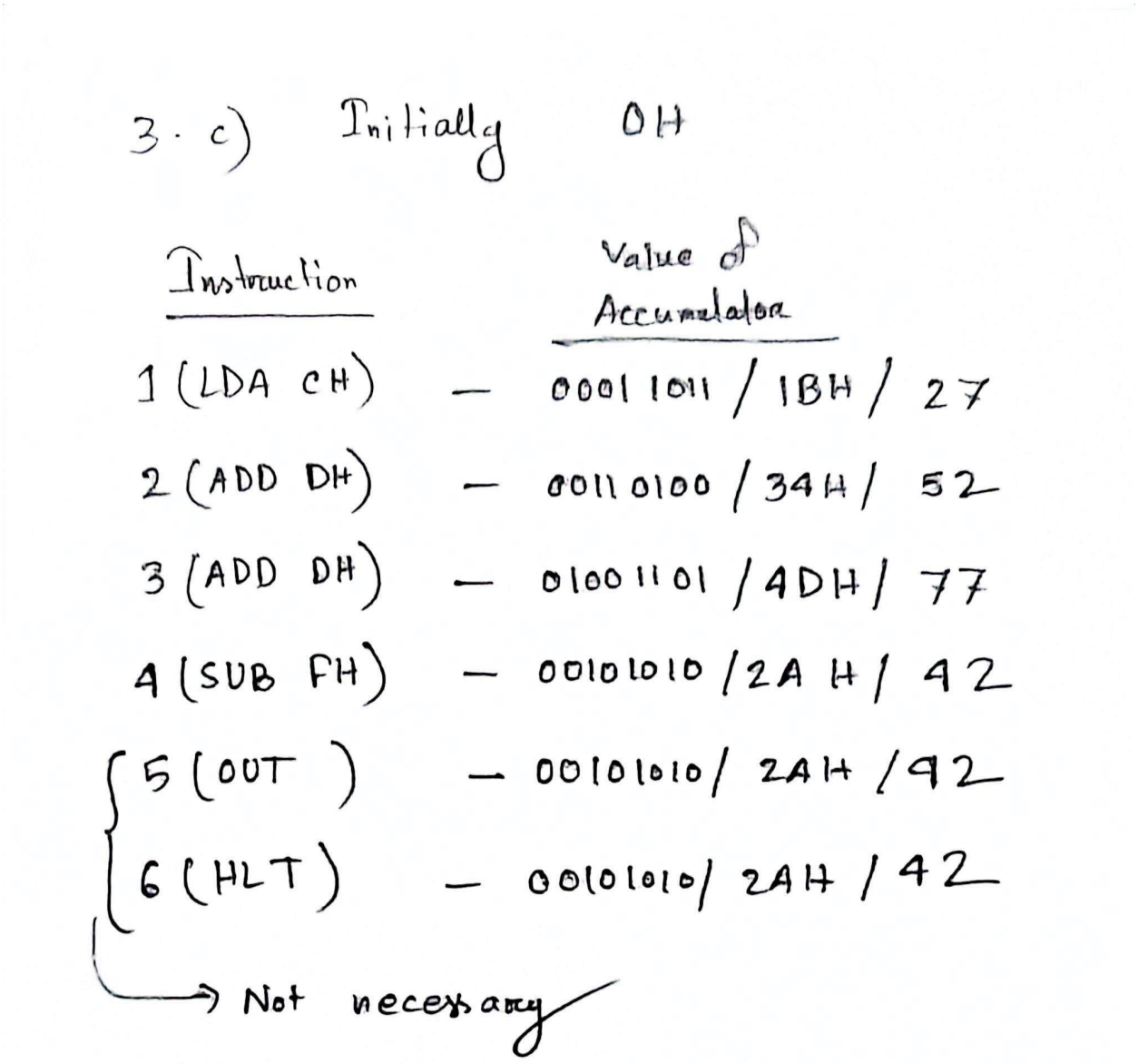


3. b)

(b)

Address	Content
0H	1100 1100
1H	0100 1101
2H	0100 1101
3H	0011 1111
4H	0001 xxxx
5H	1000 xxxx
6H	xxxx xxxx
7H	xxxx xxxx
8H	xxxx xxxx
9H	xxxx xxxx
AH	xxxx xxxx
BH	0110 0100
CH	0001 1011
DH	0001 1001
EH	0001 0100
FHH	0010 0011

3. c)



Note: You can write the value of the accumulator register in any number format(binary/decimal/hexadecimal) unless mentioned.

4.

4. Firstly, define a new opcode for LDBout in SAP-1 architecture. Then, fetch the value normally like other instruction in T_1, T_2, T_3. Now, for T_4:

CON = 0001 1010 0011

= 1A3H [Load value in the RAM ~~B register~~]

T_5:

CON = 0010 1110 0001

= 2E1H [Load the value in B register]

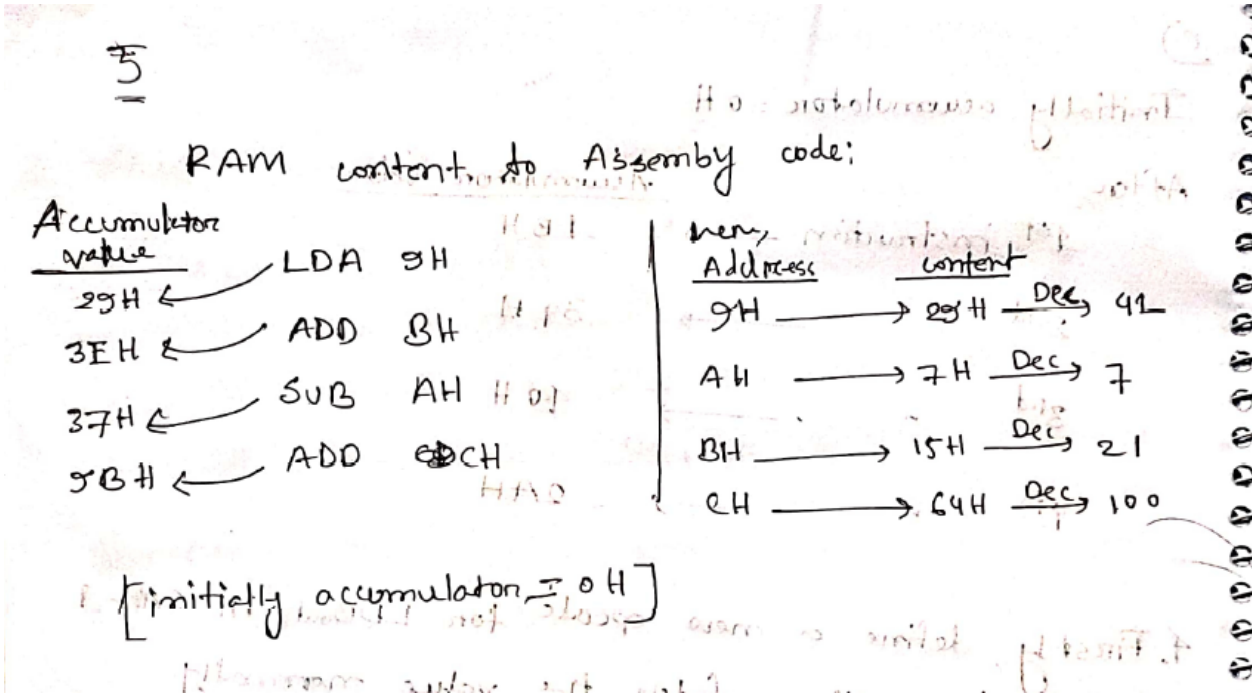
T_6:

CON = 0011 1110 0110

= 3E6 [Display the value of B to binary display]

[here initially accumulator is 0H]

5.



Note: You can write the value of the accumulator register in any number format(binary/decimal/hexadecimal) unless mentioned.

6.

